Signals & Systems (EE504)
Pre-Reading Material – Lecture 4
Characterization of Signals
Estimated Reading Time: 20–25 minutes

Course Context

This lecture explores the various ways to characterize and classify systems based on their physical behavior and mathematical properties. Building on the understanding of signals, we now examine how systems transform these signals. The lecture covers the distinction between **Single-Input Single-Output (SISO)** and **Multiple-Input Multiple-Output (MIMO)** systems, along with important system properties such as **memory, invertibility, causality, stability, time-invariance, and linearity**.

Why This Topic Matters

In engineering, a system is an interconnection of components designed to produce a desired output from a given input. Understanding properties such as **causality** and **stability** is essential for determining whether a system can be realized in real time and whether it will operate safely under different conditions. Furthermore, **linearity** and **time-invariance** enable engineers to apply powerful mathematical tools that simplify the analysis and design of complex real-world systems, including industrial processes, communication systems, and signal processing applications.

Learning Outcomes

After completing this lecture, you should be able to:

- Distinguish between **SISO** and **MIMO** systems and represent them using vector notation.
- Classify systems as **memoryless (static)** or **dynamic (with memory)**.
- Determine whether a system is **invertible** and identify its corresponding inverse system.
- Define **causality** and explain why real-time systems must be non-anticipatory.
- Assess system **stability** using the **Bounded-Input Bounded-Output (BIBO)** criterion.
- Test whether a system is **time-invariant**.
- Verify the **linearity** of a system using the principles of superposition and homogeneity.

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Activate Your Prior Knowledge

Recall from previous lectures:

- A **continuous-time system** processes signals such as (x(t)) to produce an output (y(t)), whereas a **discrete-time system** processes sequences such as (x[n]) to produce (y[n]).
- **Hybrid systems**, such as **Analog-to-Digital (A/D)** and **Digital-to-Analog (D/A)** converters, bridge the continuous-time and discrete-time domains.
- Basic components such as **resistors** are instantaneous (static), whereas **capacitors** and **inductors** involve differential equations and initial conditions, indicating that they "remember" past states.

Prerequisite Concepts

- **Vector Notation:** A column vector used to represent multiple inputs or outputs in MIMO systems.
- **Transformation ((T) or (S))**: The mathematical operator that describes how a system transforms an input signal into an output signal.
- **Identity System:** A system whose output is identical to its input. It serves as the reference model for understanding inverse systems.

Core Concepts

1. Memory in Systems

Memoryless (Static) Systems

The present output (y(t)) depends only on the present input (x(t)).

Example: A resistor.

Dynamic (Systems with Memory)

The output depends on past inputs, past outputs, or internal stored energy. Systems containing energy-storage elements such as capacitors and inductors are dynamic because their behavior depends on initial conditions.

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2. Invertibility

A system is **invertible** if distinct inputs always produce distinct outputs. In such cases, an **inverse system** can be connected in cascade with the original system to recover the original input signal.

Example

- **Multiplier:** (y(t)=kx(t)) is invertible (provided (k).
- **Squarer:** (y(t)=x^2(t)) is **not invertible** because both positive and negative input values produce the same output.

3. Causality (The Real-Time Constraint)

Causal System

A system is **causal** (or **non-anticipatory**) if its present output depends only on present and past inputs.

Non-Causal System

A **non-causal** system depends on future input values. Such systems cannot be implemented in real-time applications because future inputs are unavailable. However, they are useful in **offline (non-real-time)** processing of previously recorded data, such as satellite image processing or post-processing of recorded signals.

4. BIBO Stability

A system is **Bounded-Input Bounded-Output (BIBO) Stable** if every bounded input produces a bounded output.

Stability is an inherent property of the system. If **even one** bounded input produces an unbounded output (for example, (y(t)=e^t) as (t tends to infinity)), the system is classified as **unstable**.

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5. Time-Invariance

A system is **time-invariant** if delaying (or advancing) the input signal results in an identical delay (or advance) in the output signal.

In other words, the system's behavior does not change with time; only the timing of the response changes.

6. Linearity

A system is **linear** if and only if it satisfies both of the following properties:

1. **Superposition:** The response to the sum of multiple inputs equals the sum of the individual responses.
2. **Homogeneity (Scaling):** Scaling the input by a constant (\alpha) scales the output by the same constant.

Note: A system described by

[y(t)=mx(t)+c]

is **nonlinear** whenever (c is not equal to 0), because a zero input does not produce a zero output.

Guided Reading Questions

1. Why is the system defined by (y(t)=x(t^2)) considered to possess memory?
2. How does the "honesty" analogy help explain the concepts of causality and stability?
3. What is the mathematical condition for a composite system (System followed by its Inverse System) to behave as an identity system?
4. Can a non-causal system be physically realized in a real-time application? Why or why not?
5. Is a signal generator or oscillator generally considered a stable or an unstable system?

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Reflection Activity

Think About a Weather Forecasting System

1. Is a weather forecasting system more likely to be a **SISO** or a **MIMO** system? What could be its possible inputs?
2. If you are analyzing recorded seismograph data collected one month ago, can your analysis system be **non-causal**? Explain your answer.

Self-Assessment

True or False

1. A system is linear if it satisfies only the principle of superposition.

Answer: False

A linear system must satisfy both the principles of superposition and homogeneity.

2. All real-time engineering systems must be causal.

Answer: True

Real-time systems cannot depend on future input values.

Short Answer

What is the difference between a memoryless system and a dynamic system?

Answer:

A **memoryless (static)** system produces an output that depends only on the present input. A **dynamic (memory)** system produces an output that depends on past (and sometimes future) inputs or outputs and often requires initial conditions for its analysis.

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Think About It

If you test a system using **1,000 different bounded inputs**, and every test produces a bounded output, can you conclusively prove that the system is stable?

Answer: No.

Similar to the analogy of testing a person's honesty, stability cannot be proven through a finite number of successful tests. However, **a single example** in which a bounded input produces an unbounded output is sufficient to conclude that the system is **unstable**.

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